
PROJECT REPORT

Telecom Customer Churn Prediction Using Machine Learning

Submitted By

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Abstract

Customer churn is one of the major challenges faced by telecommunication companies. Customer churn refers to customers discontinuing their services and switching to another provider. Predicting churn helps companies identify customers who are likely to leave so that they can take preventive actions.

This project uses Machine Learning to predict whether a customer will churn based on demographic information, account details, and service usage. The Logistic Regression algorithm is used to classify customers into churn and non-churn categories. Data preprocessing techniques such as handling missing values, encoding categorical variables, and feature scaling are applied before training the model. The trained model achieves good prediction accuracy and can help telecom companies improve customer retention strategies.

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1. Introduction

Customer retention is more cost-effective than acquiring new customers. Telecom companies collect large amounts of customer information that can be analyzed to identify customers who are likely to discontinue their services.

Machine Learning algorithms can learn patterns from historical customer data and accurately predict customer churn. This project develops a predictive model that helps telecom companies identify high-risk customers and improve customer satisfaction.

2. Problem Statement

The objective of this project is to predict whether a telecom customer will leave the company based on customer information, subscription details, and service usage.

3. Objectives

The main objectives of this project are:

- To analyze telecom customer data.
 - To preprocess the dataset.
 - To train a Machine Learning model.
 - To predict customer churn.
 - To evaluate model performance.
 - To save the trained model for future predictions.
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4. Dataset Description

The dataset contains customer information collected by a telecom company.

Features include:

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure

- Phone Service
- Internet Service
- Contract Type
- Monthly Charges
- Total Charges
- Payment Method
- Churn Label (Target Variable)

Target Variable

Churn Label

- 0 → Customer will not churn
- 1 → Customer will churn

5. Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Manipulation
NumPy	Numerical Computing
Matplotlib	Data Visualization
Seaborn	Visualization
Scikit-learn	Machine Learning
Jupyter Notebook	Development Environment
Joblib	Model Saving

6. Methodology

The project follows these steps:

1. Import libraries
2. Load dataset
3. Perform exploratory data analysis
4. Handle missing values
5. Encode categorical variables

6. Split data into training and testing sets
 7. Scale numerical features
 8. Train Logistic Regression model
 9. Predict customer churn
 10. Evaluate model performance
 11. Save trained model
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7. Data Preprocessing

The following preprocessing techniques were applied:

- Removed unnecessary columns such as CustomerID.
 - Handled missing values.
 - Converted categorical variables into numerical values using Label Encoding.
 - Split the dataset into training and testing data.
 - Applied StandardScaler to normalize feature values.
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8. Model Development

The project uses the **Logistic Regression** algorithm because it is efficient, easy to understand, and performs well on binary classification problems.

Steps:

- Import Logistic Regression
 - Train the model using training data
 - Predict customer churn using testing data
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9. Model Evaluation

The model performance was evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Precision
- Recall
- F1 Score

These metrics provide a comprehensive evaluation of the classification model.

10. Results

The Logistic Regression model successfully predicts customer churn.

Sample Result

Metric	Value
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Accuracy	79.35%
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Model	Logistic Regression
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Prediction	Churn / Not Churn
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The model provides reliable predictions that can support customer retention decisions.

11. Advantages

- Easy to understand
 - Fast prediction
 - Low computational cost
 - Helps reduce customer loss
 - Supports business decision-making
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12. Limitations

- Accuracy depends on data quality.
 - Logistic Regression assumes a linear relationship between features and the target.
 - The model may not capture highly complex patterns.
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13. Future Scope

The project can be improved by:

- Using Random Forest, XGBoost, or LightGBM for higher accuracy.
- Building a web application using Flask or FastAPI.
- Deploying the model on cloud platforms.
- Performing real-time churn prediction.
- Adding dashboards for business analytics.

Conclusion

This project demonstrates the application of Machine Learning in predicting telecom customer churn. The Logistic Regression model was trained using customer data and achieved satisfactory performance. The developed system can assist telecom companies in identifying customers who are likely to leave, allowing them to implement effective retention strategies. The project also provides a foundation for deploying predictive analytics solutions in real-world business environments.

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